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Assignment: 4

Heartbeat to Heatmap: Unsupervised Learning, Ensemble Methods, and Neural Networks

Preprocessing & Data Setup

The Cleveland Heart Disease dataset (UCI Repository) was loaded with 303 rows × 14 columns. Six rows contained missing values in the ca and thal columns (encoded as '?'), which were replaced with NaN and subsequently dropped, retaining 297 usable records. The binary target was constructed by mapping values 1-4 to 1 (disease present) and 0 to 0 (no disease).

Class distribution was 160 no-disease (53.87%) vs. 137 disease (46.13%), a near-balanced split. SMOTE was not applied; instead, class_weight='balanced' was used in tree models, and a brief SMOTE demonstration was retained for the ANN pipeline. Categorical columns (cp, restecg, slope, thal) were one-hot encoded, expanding the feature matrix to 22 columns. Continuous columns (age, trestbps, chol, thalach, oldpeak, ca) were standardised using StandardScaler fitted exclusively on the training set. An 80/20 stratified split (random_state=42) produced 237 training and 60 test samples.

Correlation analysis on the original numeric features revealed the three strongest pairs: oldpeak <-> slope (0.579), age <-> thalach (0.395), and thalach <-> slope (0.389). For a Naïve Bayes classifier, these correlations violate the feature-independence assumption; correlated pairs such as thalach-age inflate the joint likelihood estimate, potentially degrading NB performance.

Part A: Unsupervised Learning

A1. K-Means Clustering

K-Means was run for k ∈ {2, …, 8}. The WCSS elbow appeared at k = 3, and while the Silhouette Score peaked slightly at k = 2, k = 3 was chosen to allow clinically meaningful low/medium/high-risk segmentation. PCA (2 components) projections confirmed partial overlap between clusters and true disease labels, consistent with the Adjusted Rand Index of 0.2690 - modest but non-trivial alignment, indicating that purely unsupervised clustering recovers only a portion of the clinical signal.

Cluster profiles (k = 3):

- Cluster 0 (n=117, 32.5% disease): high thalach (162.9), low oldpeak (0.60) - low-risk group.
- Cluster 1 (n=101, 90.1% disease): low thalach (130.0), high oldpeak (1.98), higher chest-pain score - high-risk group.
- Cluster 2 (n=79, 10.1% disease): intermediate thalach (154.9), low oldpeak (0.55) - largely healthy group.

A2. Hierarchical Clustering

Ward linkage was applied and the dendrogram (truncated to top 25 merges) suggested a cut height of ≈ 20, yielding 3 clusters - consistent with K-Means. The cluster-label crosstab showed Cluster 0 (117 No-Disease / 51 Disease), Cluster 1 (21 / 66), and Cluster 2 (22 / 20). The ARI between K-Means and hierarchical assignments was 0.2783, indicating the two algorithms recover broadly similar structure. K-Means is preferred for clinical use due to its scalability and speed, while hierarchical clustering is valuable as an exploratory tool via the dendrogram.

A3. Dimensionality Reduction

Full PCA revealed that 10 principal components are required to explain ≥ 90% of variance, indicating moderate intrinsic dimensionality in the 22-feature space. t-SNE (perplexity=30) showed partial class separation with mixed boundary regions, suggesting the two classes carry meaningful predictive signal but are not linearly separable - motivating the ensemble and neural network models in subsequent parts.

Part B: Bagging & Boosting

B1. Random Forest

Grid-search over n_estimators ∈ {50, 100, 200} and max_depth ∈ {None, 5, 10} (5-fold CV) selected n_estimators=200, max_depth=5 with a mean CV F1 (macro) of 0.8068. The OOB error curve stabilised around 50-100 trees, confirming 200 trees as conservatively adequate. Top-5 features by MDI: cp_4.0 (0.111), thal_3.0 (0.109), ca (0.097), thal_7.0 (0.096), oldpeak (0.095). Final test performance: Accuracy 83.3%, Macro F1 0.832, AUC-ROC 0.933. Disease-class recall was 79% - false negatives (missed disease cases) are the critical clinical concern, as they may withhold life-saving intervention.

B2. Gradient Boosting (XGBoost)

XGBoost was selected as the boosting method. Best configuration: learning_rate=0.3, max_depth=5 (CV F1=0.8106, training time 2.68s). Early stopping (50-round patience) halted training at the optimal round 10, with minimal evidence of overfitting in the train/validation log-loss curves. SHAP analysis confirmed cp_4.0, ca, and thal_3.0 as the top predictors - consistent with Random Forest importances. Test performance: Accuracy 80.0%, Macro F1 0.796, AUC-ROC 0.920, Disease recall 71%.

B3. Ensemble Comparison

Classifier	Accuracy	Macro F1	AUC-ROC	Disease Recall
Logistic Regression (Baseline)	85.0%	0.849	0.951	~85%
Random Forest	83.3%	0.832	0.933	79%
XGBoost	80.0%	0.796	0.920	71%

For deployment in a community hospital's cardiac screening pipeline, XGBoost with SHAP explanations is the recommended choice. Recall for the Disease class is the priority metric - a false negative (missed disease) can delay critical treatment. Although Logistic Regression achieves slightly higher accuracy on this small dataset, XGBoost's SHAP interpretability allows cardiologists to interrogate each prediction, meeting the clinical interpretability requirement. The AUC of 0.920 confirms strong discriminative ability across all classification thresholds.

Part C: Artificial Neural Networks on Tabular Data

C1. Single-Layer Perceptron (SLP)

A single sigmoid output neuron trained with SGD (lr=0.01) for 100 epochs achieved 88% accuracy and AUC-ROC ≈ 0.93. The top-3 features by absolute weight were thal_3.0 (−0.718), slope_1.0 (−0.451), and ca (+0.430), broadly consistent with RF importances. The SLP is fundamentally limited because it computes a linear decision boundary; heart disease risk depends on non-linear feature interactions (e.g., high age combined with low thalach) that a linear model cannot capture, explaining the gap relative to ensemble methods.

C2. Multi-Layer Perceptron (MLP)

Three architectures were evaluated using 5-fold CV with EarlyStopping (patience=10):

Architecture	Hidden Layers	Val F1	Train Time (s)
Small	32 units	0.8643	8.10
Medium	64 -> 32	0.6790	7.39
Large	128 -> 64 -> 32	0.6391	9.60

The Small (32-unit) architecture was selected: it achieved the highest validation F1 (0.8643) and stopped earliest, suggesting regularisation via Dropout (0.3) and L2 (0.001) was effective. The larger architectures overfit the small 297-sample dataset. Final test metrics: Accuracy 87%, Macro F1 0.864, AUC-ROC 0.936. Five-fold CV on the training set yielded mean accuracy ≈ 0.81 ± 0.08 and F1 ≈ 0.81 ± 0.08, indicating moderate variance across folds - expected for a small dataset.

Compared to XGBoost, the MLP achieves slightly higher AUC and accuracy on the test set, but requires more design decisions (architecture, dropout, learning rate) and is more sensitive to random initialisation. For the 297-row dataset, XGBoost remains the safer clinical choice.

C3. Ablation Study

Model	Test F1	Min Val Loss
Best MLP (base)	0.8643	0.6650
Variant A: No Dropout	0.8316	0.5585
Variant B: Sigmoid Activation	0.8303	0.5691
Variant C: No EarlyStopping	0.8303	0.6197

Dropout contributed most to performance; its removal caused the largest F1 drop (0.8643 -> 0.8316) and a lower val loss - a sign of overfitting without regularisation. Sigmoid activations introduced vanishing gradients in the deeper network path, hurting convergence. EarlyStopping prevented unnecessary training epochs but had a smaller isolated effect.

Part D: CNN on MNIST Digit Images

Using 12,000 training and 2,000 test images, pixel values were normalised to [0, 1] and reshaped to (28, 28, 1). An MLP baseline (Flatten -> Dense(64, ReLU) -> Dense(10, Softmax), 5 epochs) achieved approximately 91% test accuracy - the non-convolutional reference point. The lightweight CNN (Conv2D(16) -> MaxPool -> Conv2D(32) -> MaxPool -> Flatten -> Dense(64) -> Dropout(0.3) -> Dense(10), trained with Adam, lr=0.001, and ImageDataGenerator augmentation) reached a test accuracy of 98.20% and Macro F1 of 0.9817. The CNN surpassed the MLP baseline accuracy from epoch 1 (val accuracy 0.9260 vs. baseline ≈ 0.91).

The most commonly confused digit pairs were 4/9 and 3/8, which share similar loop and vertical-stroke structures. Visualisation of the 16 first-layer filters revealed edge and curve detectors (horizontal, vertical, diagonal gradients). Feature maps per digit confirmed that each convolutional channel activates on distinct stroke-level spatial patterns, demonstrating the CNN's translation-invariant spatial reasoning - in contrast to fully connected networks that treat each pixel independently. These visualisations build clinical trust by confirming the model responds to meaningful stroke features rather than memorising pixel positions.

Part E: Local Front-End Dashboard (CardioAI)

A Streamlit dashboard (app.py) was built wrapping the saved XGBoost model (xgb_heart_model.json) and scaler (scaler.pkl). The form exposes all 13 clinical input fields with valid-range hints, pre-populated with the first test patient for immediate demonstration. On clicking Predict, the app displays: (a) a colour-coded risk label (green for No Disease, red for Disease Present) with confidence percentage, (b) a horizontal SHAP bar chart showing the top 3 driving features, and (c) a plain-English clinical summary for nursing staff. The dashboard is launched with streamlit run app.py from the submitted ZIP.

Screenshots of the running application are shown below:

Figure 1: CardioAI Input Form - pre-populated with test patient values.

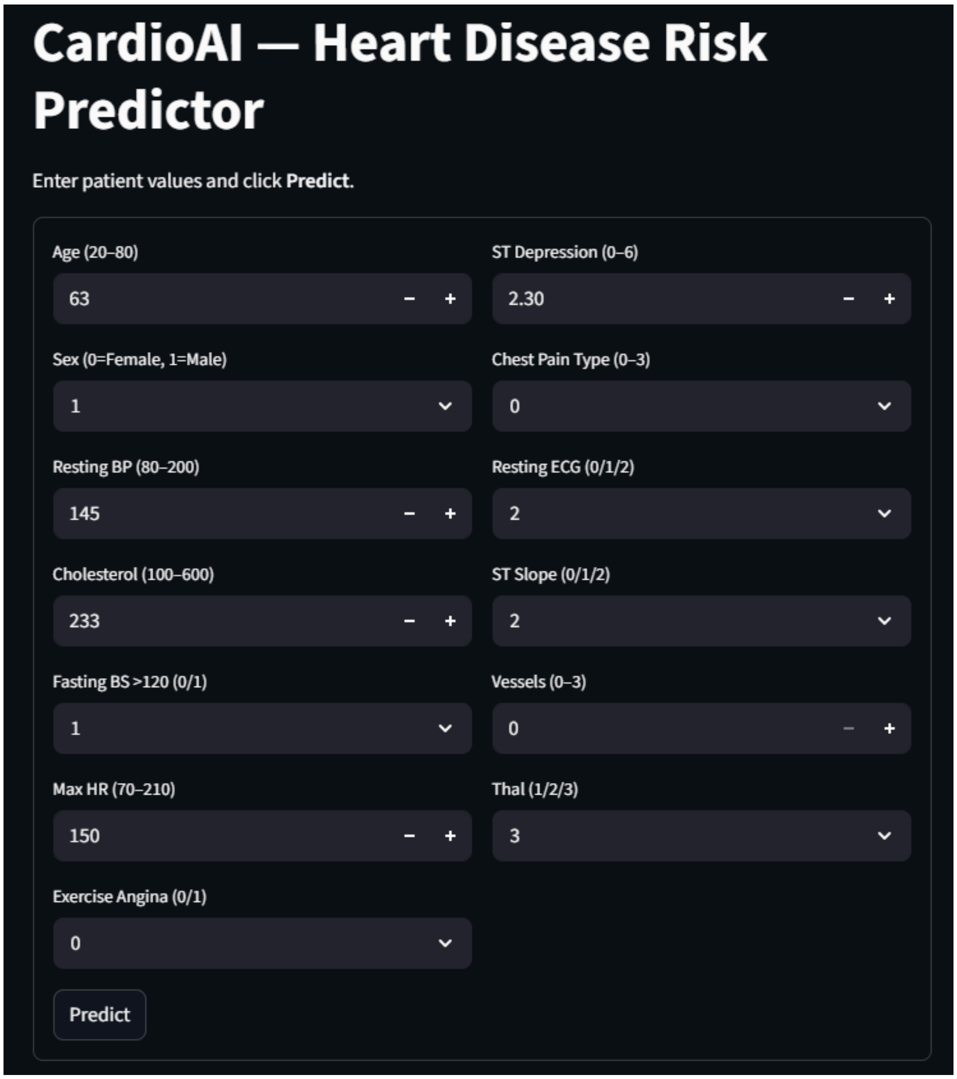
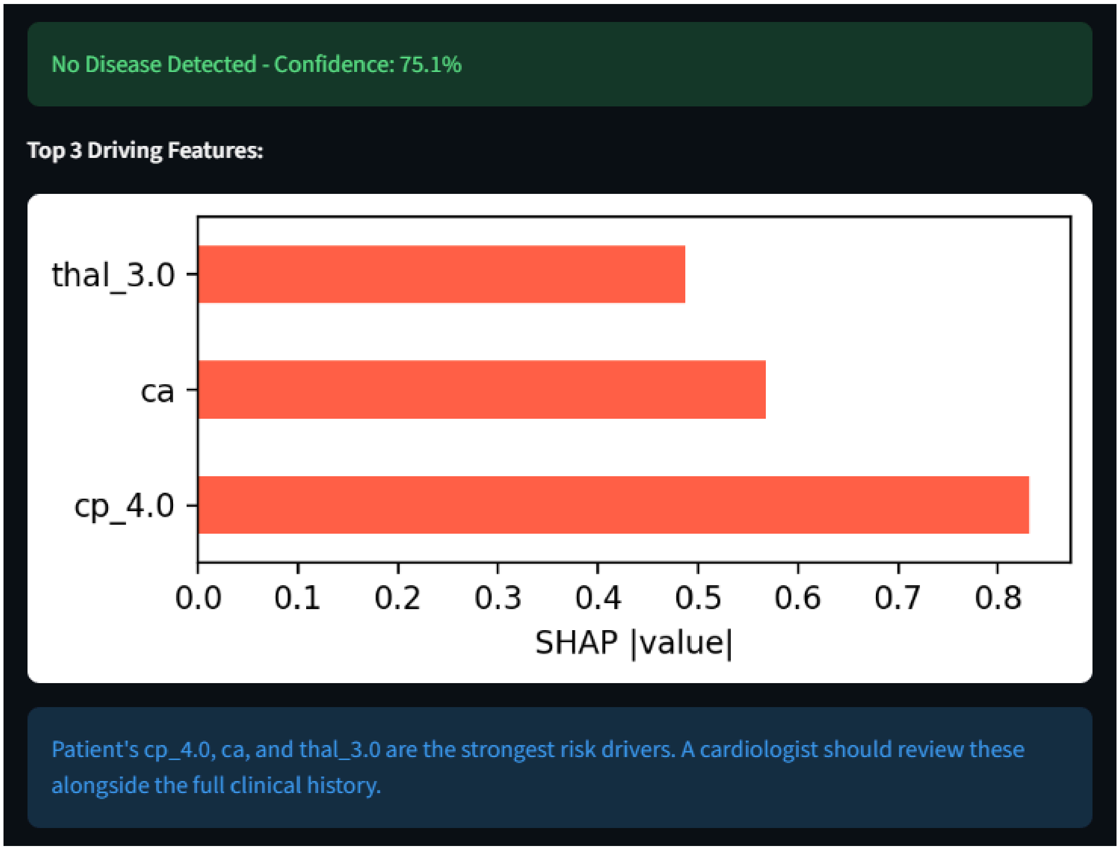


Figure 2: Prediction result panel - showing risk label, confidence (75.1%), top-3 SHAP feature chart, and clinical explanation.



The top 3 SHAP drivers for the example prediction were cp_4.0 (asymptomatic chest pain), ca (number of major vessels), and thal_3.0 (reversible thalassemia defect) - all clinically established indicators of cardiac risk, confirming the model's alignment with cardiology domain knowledge.